MILITARY SPECIFICATION

FUZE, XM596; SETBACK PIN AND HOUSING ASSEMBLY FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-F-50421 (MU), dated 5 May 1969 and is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

Page 1

Add the following Preamble underneath the title:

"This specification is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense."

Page 2

Add new paragraph 3.2.1 to read as follows:

"3.2.1 Qualification test for M55 detonator. The detonator shall function the test cartridge high order when tested as specified in 4.4.15."

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Add new paragraph 3.5.1 to read as follows:

"3.5.1 Safety. There shall be no premature burst (see 6.7) or evidence thereof in any of the test firings when tested as specified in 4.4.12."

3.7: Delete

"Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined."

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4.2.2.1.9: Delete in its entirety and substitute the following:

"4.2.2.1.9 Jolt. Twelve (12) setback pin and housing assemblies shall be assembled to M533 fuzes and subjected to this test."

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4.2.2.1.10: Delete in its entirety and substitute the following:

"4.2.2.1.10 Jumble. Twelve (12) setback pin and housing assemblies shall be assembled to M533 fuzes and subjected to this test."

Add new paragraph 4.4.2.1.11 to read as follows:

"4.2.2.1.11 Hardness of verge shaft and pinion shaft. Fifty (50) verge shafts and fifty (50) pinion shafts shall be subjected to this test."

4.2.4.2:

Delete: "After completion of tests specified in 4.2.4.1, the three-hundred (300) assemblies shall be subjected to this test" and substitute "After completion of test specified in 4.2.4.1, the three-hundred (300) assemblies shall be removed from their regular shipping containers, assemble to M533 fuzes, and subjected to this test."

Second sentence, after requirements: Add:

"or one (1) premature burst occurs."

4.2.4.3:

Delete: "Ten (10) setback pin and housing assemblies shall be subjected to this test" and substitute "Ten (10) setback pin and housing assemblies shall be assembled to M533 fuzes and subjected to this test."

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4.3.2.3: Delete in its entirety.

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4.3.2.4: Delete in its entirety.

4.3.2.5: Delete in its entirety.

4.3.2.6: Delete in its entirety.

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4.3.2.7: Delete in its entirety.

4.3.2.8: Delete in its entirety.

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Add new paragraph 4.3.2.14 to read as follows:

"4.3.2.14. Housing Assembly (See Dwg. 8876220 covering a detail of Dwg. 8880916).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65%		
101.	Largest outside diameter	Gage	28001
102.	Diameter of dowels	Gage	28002
103.	Height of dowels	Gage	03003
104.	Outside diameter of lower shaft plate	Gage	04001
Minor	AQL 0.65%		
201.	Evidence of poor workmanship (See (3.7)	Visual	02003"

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4.3.3.2, last sentence: Delete

"Assembly" and substitute "assembling."

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4.3.3.15:

Delete: "(see 3.6)" and substitute "(see 3.5)."

Delete: "(After completion of test specified in 4.3.3.14, two-hundred (200) assemblies shall be subjected to this test." and substitute "After completion of test specified in 4.3.3.14, two-hundred (200) assemblies shall be removed from their shipping containers. Assemble to M533 fuzes, and subjected to this test."

4.3.3.16:

Delete: "(See 3.7)" and substitute "(See 3.6)."

Delete: "Ten (10) additional assemblies shall be subjected from each lot and subjected to this test." and substitute "Ten (10) additional assemblies shall be selected from each lot, assemble to M533 fuzes, and subjected to this test."

Add new paragraph 4.3.3.17 to read as follows:

"4.3.3.17 Hardness of verge shaft and pinion shaft (See Dwgs. 8876206 and 8876208), Major defect, Code No. 29001. Twenty (20) each verge shafts and pinion shafts shall be selected from each heat treat batch and subjected to this test. If one (1) or more of each twenty (20) fails to comply with the requirement specified on the applicable drawing, the batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.14."

Add new paragraph 4.3.3.18 to read as follows:

"4.3.3.18 Qualification test for M55 detonator (see 3.2.1), Major defect. One-hundred and twenty-five (125) detonators shall be randomly selected from each detonator lot

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and subjected to this test. If any detonator fails to comply with the applicable requirement, the detonator lot shall be rejected for use in setback pin and housing assemblies. Note: The detonators selected for this test shall be tested with the lead cup lot with which they will be used. Change in lead cup lot requires retest of detonators. The test shall be performed as specified in 4.4.15."

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Add new paragraph 4.4.14 to read as follows:

"4.4.14 Hardness of verge shaft and pinion shaft. The verge shaft and pinion shaft shall be subjected to the Rockwell hardness test in accordance with ASTM Test Method E 18-67. Parts so tested shall not be returned to the lot."

Add new paragraph 4.4.15 to read as follows:

"4.4.15 Qualification test for M55 detonator. The detonator and lead cups shall be assembled an M533 fuze and assembled to a test cartridge (dwg. 9231745). The test cartridge shall be conditioned at -65°F for 16 hours min and fired in accordance with 4.4.12."

5.1.1: Delete "8880918" and substitute "9247418."

5.2.1: Delete "9230368" and substitute "9247417."

5.3: Delete "8799112" and substitute "9247417, and 9247418."

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Add new paragraph 6.7 to read as follows:

"6.7 Premature burst. A premature burst is considered to have occurred if the round functions in the gun bore or in the first 18 meters of flight."

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